

Kartikeya Chitranshi

kartikeya4524@gmail.com — +49 176 61668557 — Berlin, Germany

Website: <https://kartikeya1234.github.io/>

LinkedIn: <https://www.linkedin.com/in/kartikeya-chitranshi/>

GitHub: <https://github.com/kartikeya1234>

PROFILE

ML researcher working on robust generalization and spurious correlation removal in deep models. Two years at Zuse Institute Berlin, with a publication (arXiv:2505.16583) showing that counterfactual fine-tuning of CLIP removes spurious correlations more effectively than adversarial perturbations. MSc in Scientific Computing from TU Berlin, thesis on adversarial and counterfactual robustness of multi-modal foundation models. Interested in sparse and minimum-norm adversarial attacks on multimodal models, robust evaluation of vision-language systems, and the empirical foundations of foundation-model security.

RESEARCH INTERESTS

Adversarial Attacks on Foundation Models, Sparse and Minimum-Norm Adversarial Examples, Adversarial Robustness Evaluation, Multimodal Model Security, Counterfactual Training, Trustworthy Machine Learning.

PUBLICATIONS

- **Training on Plausible Counterfactuals Removes Spurious Correlations**

Sadiku, S., **Chitranshi, K.**, Kera, H., and Pokutta, S.

arXiv preprint, arXiv:2505.16583 (2025) Zuse Institute Berlin & TU Berlin.

Demonstrated that training on plausible counterfactual explanations removes spurious correlations more effectively than adversarial perturbations, achieving 12% better worst-group accuracy over standard training on WaterBirds. Contributions: counterfactual and adversarial training pipeline implementation, CLIP fine-tuning on multi-GPU infrastructure, and evaluation across image classification and image-text retrieval benchmarks.

EDUCATION

Master: MSc. in Scientific Computing

[Apr 2022] – [Jul 2025]

TU Berlin – Berlin, Germany

Final grade: 2.0 (gut)

Supervisor: Prof. Dr. Sebastian Pokutta (TU Berlin / ZIB)

Relevant coursework: High-Dimensional Optimization and Learning, Numerical Linear Algebra, Deep Learning, Machine Intelligence, Optimization under Uncertainty.

Master Thesis: [Robustness of Multi-Modal Foundation Models](#)

[Oct 2024] – [Apr 2025]

TU Berlin – Berlin, Germany

- Investigated how sparse adversarial attacks and plausible counterfactual perturbations affect vision-language models on image captioning and visual question answering tasks, finding that imperceptible perturbations can severely degrade model outputs.
- Showed that CLIP models fine-tuned on counterfactual images achieve stronger robustness than those fine-tuned on adversarially-generated images across image classification and image-text retrieval benchmarks.
- Implemented sparse adversarial attack algorithms and counterfactual training pipelines in PyTorch and HuggingFace Transformers on multi-GPU infrastructure with SLURM, enabling systematic comparison of training paradigms at scale.
- Deployed findings as an interactive public system (robust-mmfm.com) to make robustness evaluation accessible to the broader research community.

Bachelor: BSc. in Physical Sciences
University of Delhi – Delhi, India

[Jul 2018] – [Jun 2021]

WORK EXPERIENCE

Student Research Assistant

[Sep 2023] – [Sep 2025]

Zuse Institute Berlin – Berlin, Germany

Group: Prof. Sebastian Pokutta, AI in Society and Science

- Conducted research on spurious correlation robustness in vision-language models, contributing to a publication on counterfactual training (Sadiku et al., arXiv:2505.16583) as second author. Work included systematic comparison of counterfactual versus adversarial fine-tuning paradigms on CLIP models across image classification and image-text retrieval benchmarks.
- Designed and implemented counterfactual and adversarial training pipelines for CLIP vision-language models in PyTorch and HuggingFace Transformers, running distributed experiments on multi-GPU clusters with SLURM at scale.
- Implemented sparse adversarial attack algorithms on CNNs and vision-language models, contributing to a comprehensive analysis of model vulnerability to imperceptible perturbations in image captioning and visual question answering tasks.

Machine Learning Intern

[Jun 2021] – [Aug 2021]

TestAIng.com – Bangalore, India

- Implemented adversarial attack algorithms (FGSM, PGD, C&W, DeepFool) on deep neural networks in TensorFlow — first exposure to model robustness evaluation.

Data Analyst Intern

[May 2020] – [Jul 2020]

Peacock Solar – Gurugram, India

- Built automated hyperparameter optimization pipelines using Optuna for renewable energy demand forecasting.

SKILLS

- **Programming:** Python (primary), Rust.
- **Deep Learning:** PyTorch, HuggingFace Transformers, PyTorch Lightning, TensorFlow.
- **Research Methods:** Adversarial attacks (FGSM, PGD, C&W, DeepFool, sparse attacks), counterfactual training, spurious correlation analysis, supervised fine-tuning, model robustness evaluation.
- **Architectures:** CNNs, Vision-Language Models (CLIP), Transformers, Large Language Models.
- **HPC:** SLURM, multi-GPU distributed training.
- **Languages:** English (C2), Hindi (Native), German (A2).

ACADEMIC RECOGNITION

- Received MSc admission offers from Imperial College London, University of Edinburgh, and Trinity College Dublin (2022); chose TU Berlin.

Kartikya Chitambar